

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



[Lab.1] Surface Structures/ Surface Energies	 3
[Lab.2] Electronic Structure depend on number of layers	 26
[Lab.3] Oxygen Reduction Reaction	 49
[Lab.4] CO ₂ Reduction Reaction	 82
[Lab.5] Nitrogen Reduction Reaction	 105

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

Learning Objectives:

→ Verify the CO₂ Reduction Reaction (CO₂RR) process on the electrocatalysts using density functional theory (DFT) calculations.

Materials: Pd@N4-Graphene



MatSQ Credit: ~ 52 CPU hour (\$ 13.0)

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

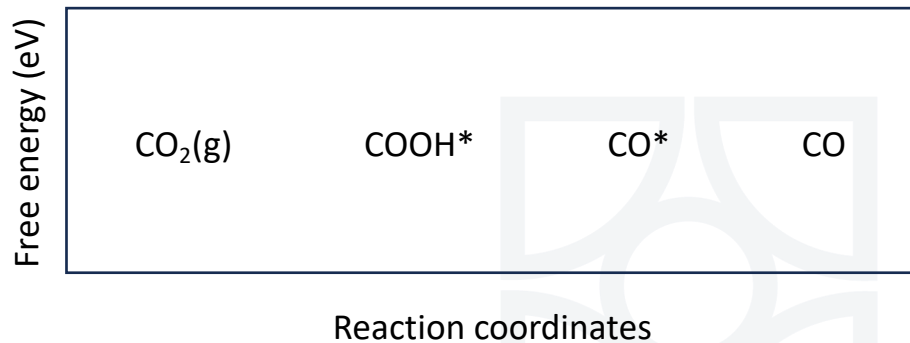
Pd@N4-Graphene

1. Build Graphene structure
2. Calculate variable cell relaxation (primitive cell)
3. Build Pd@N4-Graphene structure
4. Optimize atomic structures of Pd@N4-Gr structure
5. Build adsorption structures
6. Optimize atomic structures (adsorption structures)
7. Analyze optimized adsorption structures
8. Build molecule structures
9. Optimize molecule structures using High Throughput
10. Calculate adsorption energies
11. Calculate free-energy diagram

MatSQ Credit: ~ 52 CPU hour (\$ 13.0)

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

CO₂RR Free energy diagram



$$\Delta E_{\text{COOH}^*} = E(\text{COOH}^*/\text{substrate}) - E(\text{substrate}) - \left\{ E(\text{CO}_2) + \frac{1}{2} E(\text{H}_2) \right\}$$

$$\Delta E_{\text{CO}^*} = E(\text{CO}^*/\text{substrate}) - E(\text{substrate}) - \{ E(\text{CO}_2) - E(\text{H}_2\text{O}) + E(\text{H}_2) \}$$

List of required calculations

- Molecules : H₂, H₂O, CO₂
- Substrate: Pd@N4-graphene
- Adsorption structures: CO₂/substrate, COOH/substrate, CO/substrate

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

1. Build Graphene structure

Structure Builder

Modeling | Cell | Atom | Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Preset Structure

Bulk Graphene (Hex) ✓

Surfaces

Nano Structures

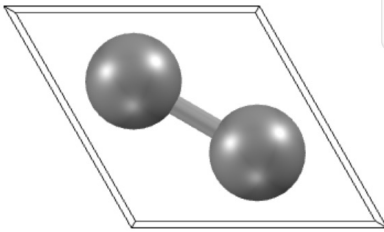
Organic molecules

Polymer

✓ Generate

HISTORY

Preset Str. ★



C: 2
a,b,c (Å): 2.46, 2.46, 20.00
α,β,γ (°): 90.00, 90.00, 119.98

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

2. Calculate variable cell relaxation (primitive cell)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic+cell optimization

☐ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☒ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

&IONS

Ion dynamics

Upscale

vc-relax

200

high

0.00038

bfgs

100

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

&CELL

Cell dynamics

Cell factor

Pressure threshold (Kbar)

Cell relax type

60

480

smearing

0.002

bfgs

2

0.2

2Dxy

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True

plain

0.5

0.000001

random

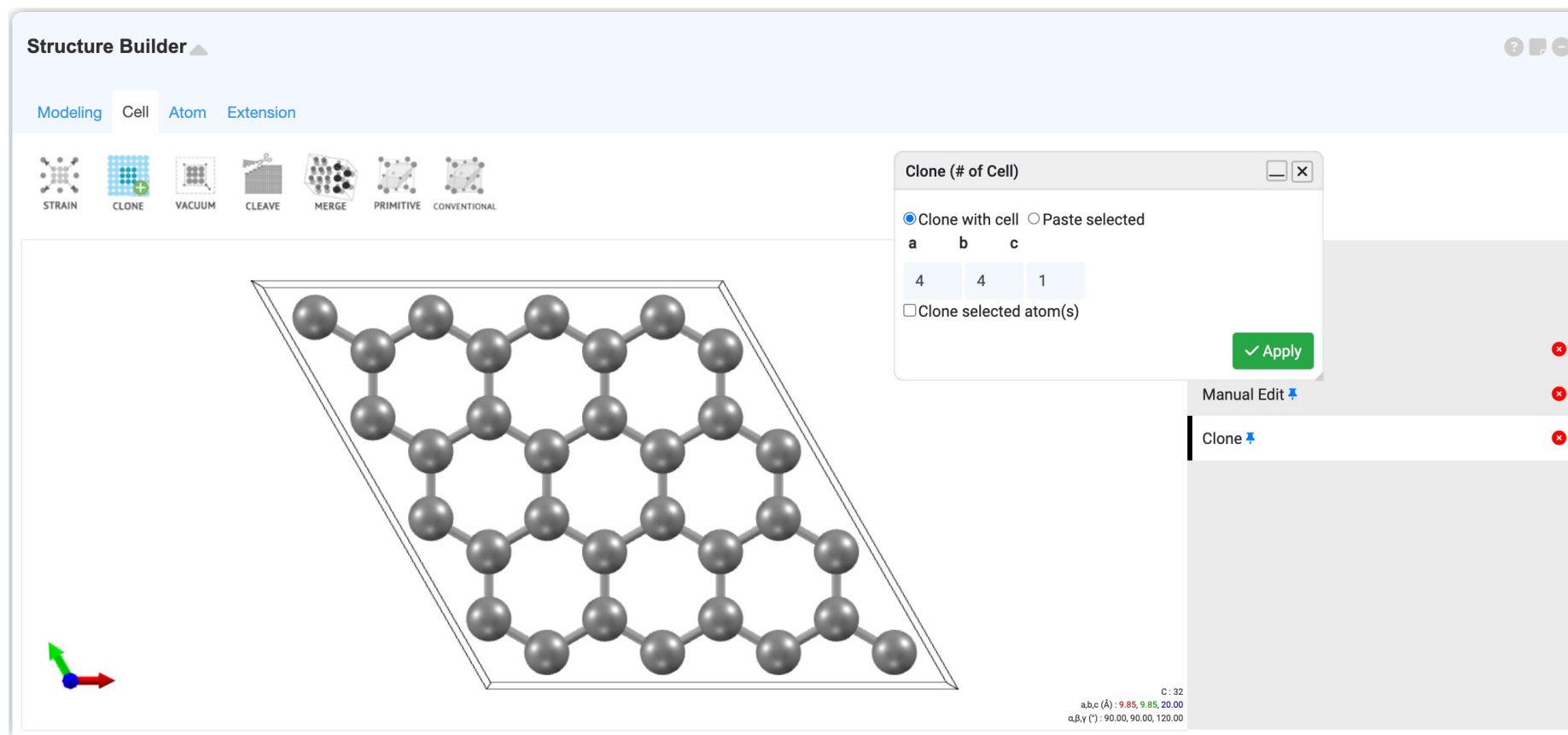
Monkhorst-Pack

12 12 1

0 0 0

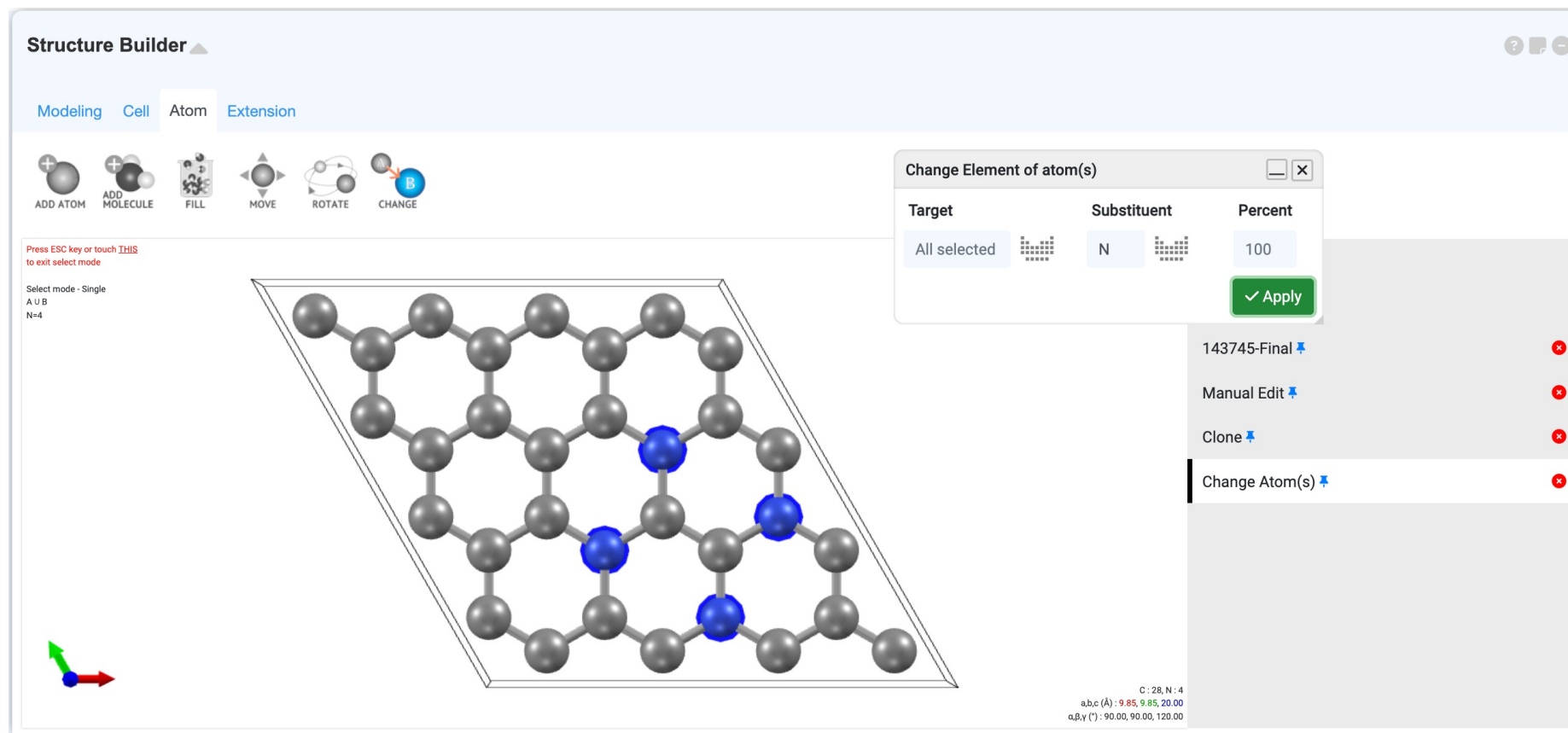
[Lab.4] CO₂ Reduction Reaction (CO₂RR)

3. Build Pd@N4-graphene structure



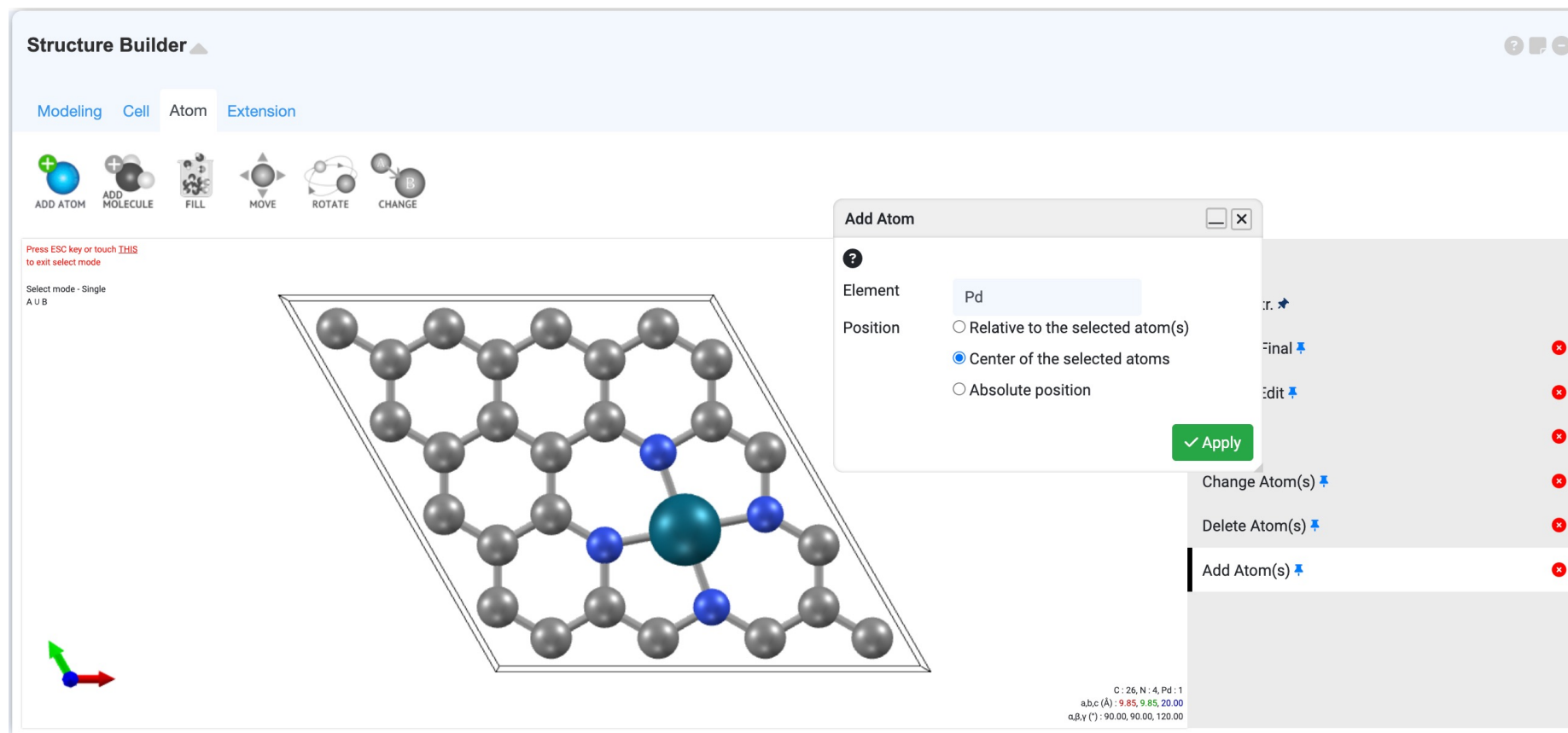
[Lab.4] CO₂ Reduction Reaction (CO₂RR)

3. Build Pd@N4-graphene structure



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

3. Build Pd@N4-graphene structure



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

4. Optimize atomic structure of Pd@N4-Gr structure

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☒ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

Number of electron spin

Starting magnetization

C

N

Pd

&IONS

Ion dynamics

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

relax

200

high

0.001

40

240

smearing

0.002

2 (up,down)

1

1

1

bfqs

200

True

plain

0.5

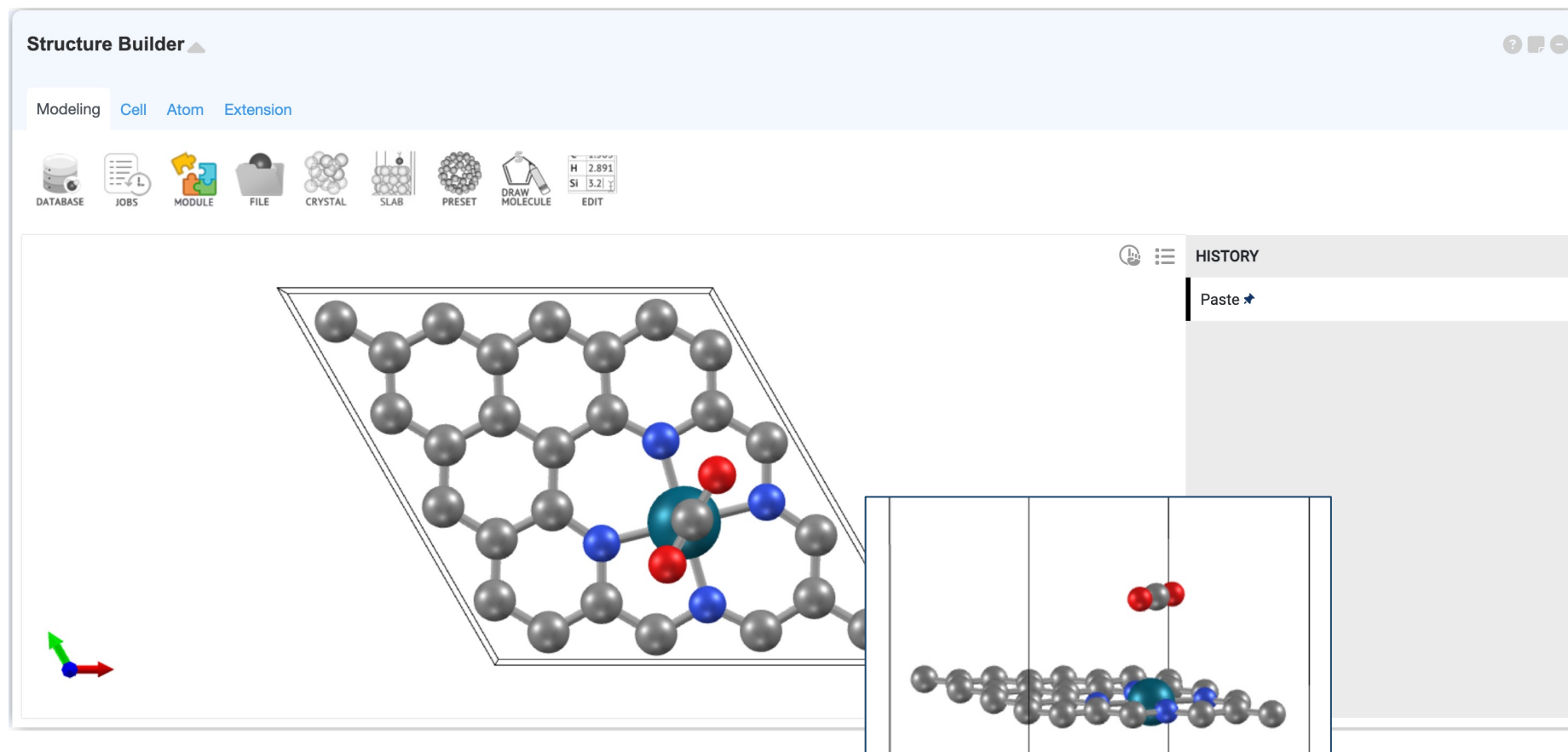
0.000001

random

Monkhorst-Pack

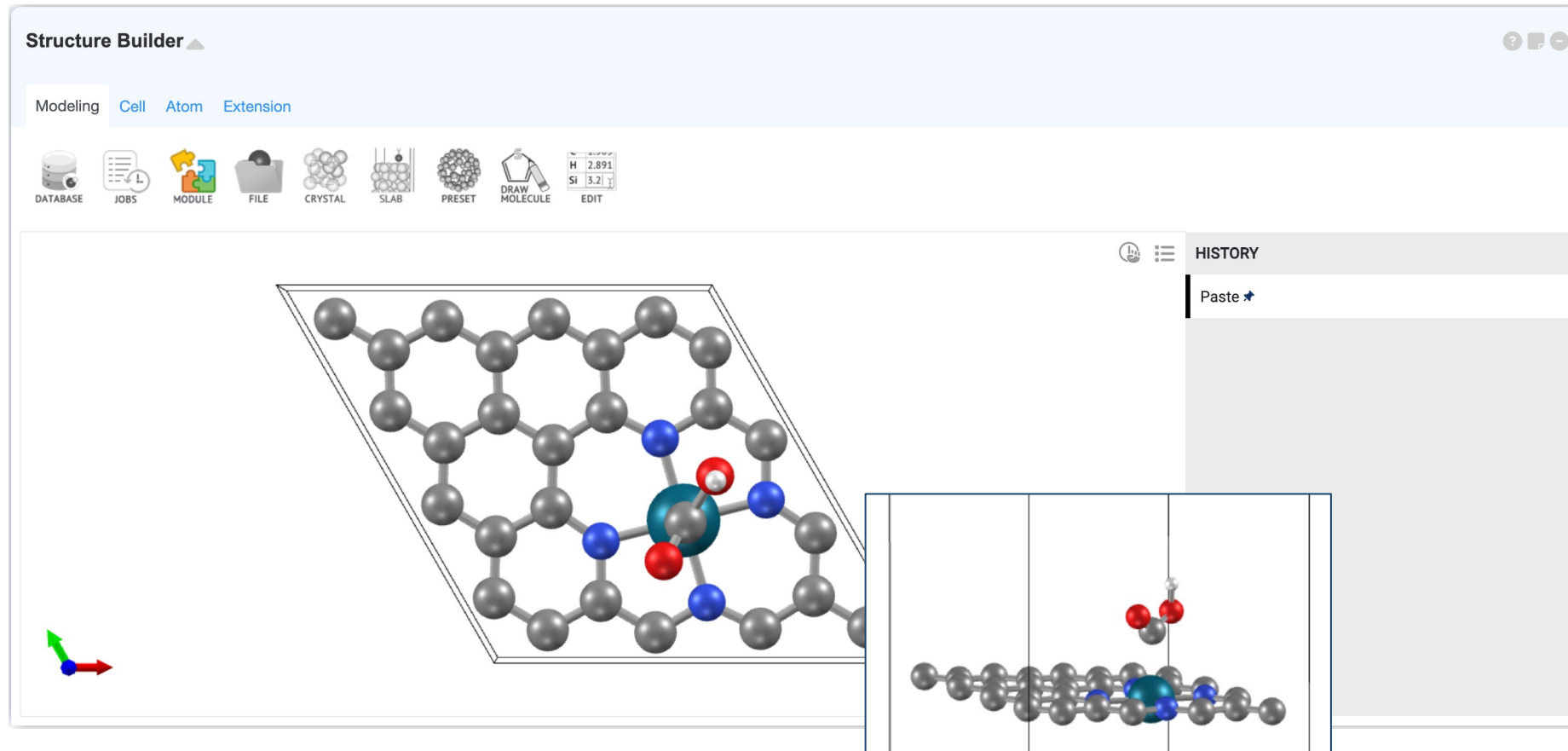
[Lab.4] CO₂ Reduction Reaction (CO₂RR)

5. Build adsorption structures (CO₂)



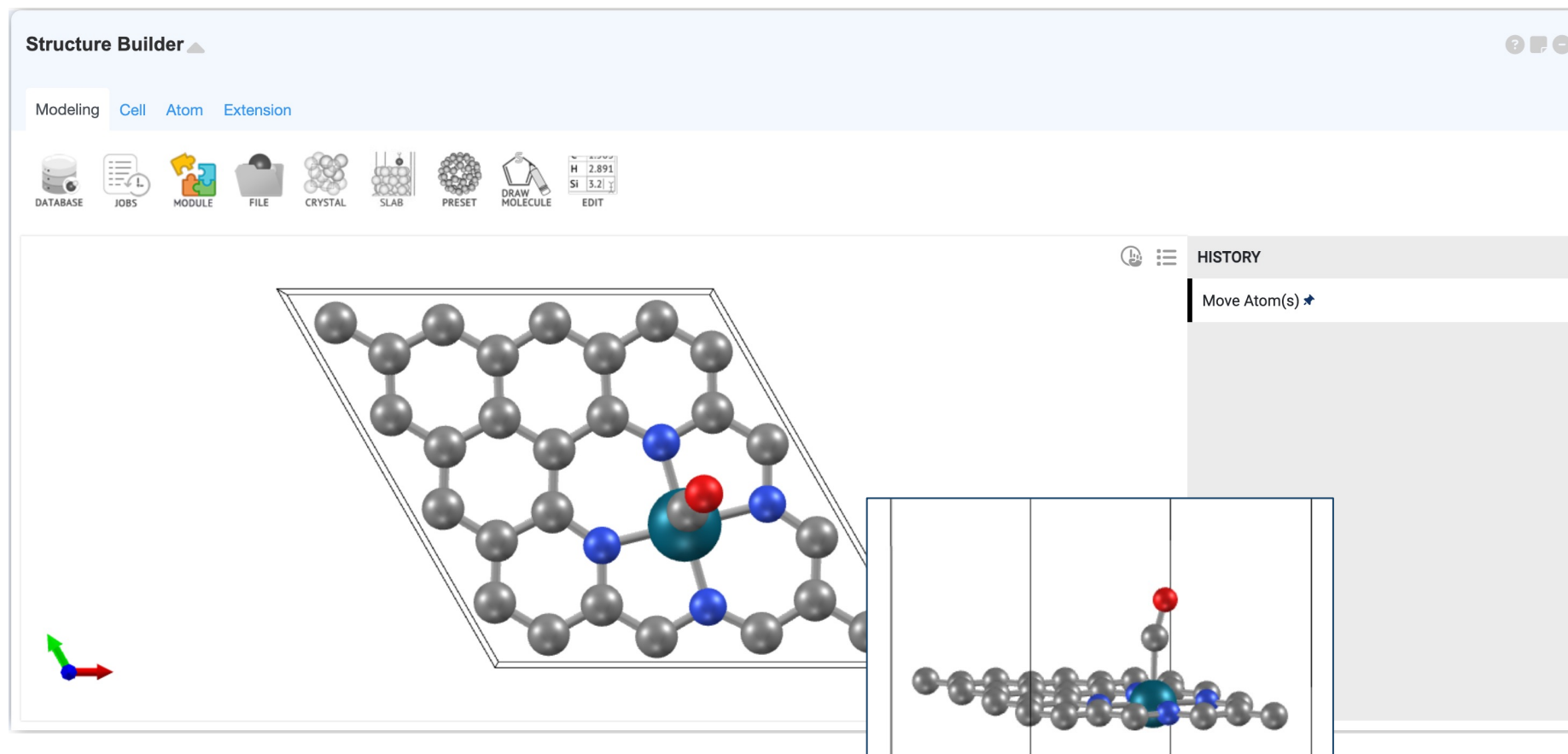
[Lab.4] CO₂ Reduction Reaction (CO₂RR)

5. Build adsorption structures (COOH*)



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

5. Build adsorption structures (CO*)



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

6. Optimize atomic structures (adsorption structures)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☒ Spin polarization
☐ Van der Waals
☐ Electric field
☐ Spin Orbit Coupling
☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

Number of electron spin

Starting magnetization

C

N

Pd

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

&IONS

Ion dynamics

KPOINTS

Sampling

relax

200

high

0.001

40

240

smearing

0.002

2 (up,down)

1

1

1

200

True

plain

0.5

0.000001

random

bfqs

Monkhorst-Pack

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

7. Analyze optimized adsorption structures



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

8. Build molecule structures

Molecules  

Modeling **Cell** Atom Extension

 DATABASE  JOBS  MODULE  FILE  CRYSTAL  SLAB  PRESET  DRAW MOLECULE  EDIT





Manual Edit 

Unit **Angstrom(Å)**

Cell **Lattice Vectors**

	a	b	c
a	15	0	0
b	0	15	0
c	0	0	15

Atoms **Cartesian** ☐ Selected Atoms

ID	EL.	x	y	z
1	H	6.75	7.5	6.75
2	H	7.95	7.5	6.75

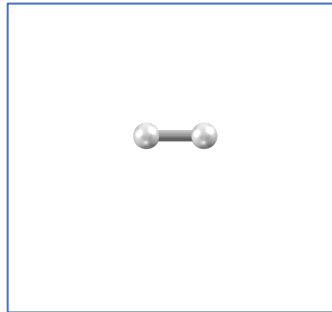
[+ Add Atom](#)

 Reset  Apply

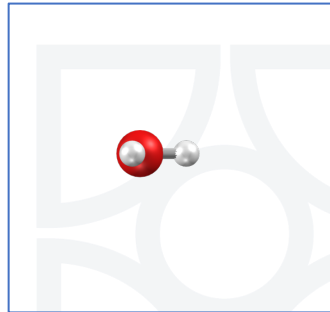
a,b,c (Å): 15.00,
α,β,γ (°): 90.00,

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

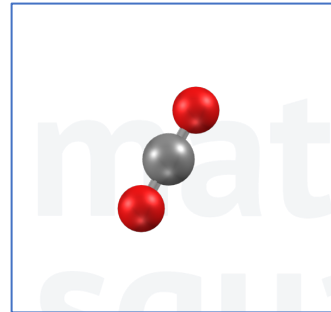
8. Build molecule structures



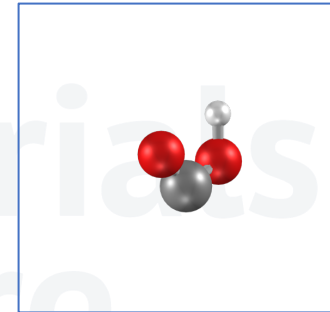
H₂



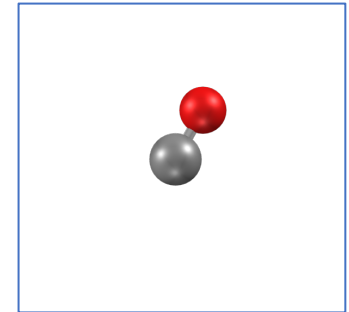
H₂O



CO₂



COOH



CO

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

9. Optimize molecule structures using High Throughput (molecules)

High Throughput ▲

Default

PWscf

```
&CONTROL
calculation = 'relax'
forc_conv_thr = 0.00038
nstep = 200
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'from_scratch'
tpnfor = .TRUE.
verbosity = 'high'
wf_collect = .TRUE.
/
```

Update

Element	Filename
H	H.pbe-kjpaw_psl.1.0.0.UPF
O	O.pbe-n-kjpaw_psl.0.1.UPF
C	C.pbe-n-kjpaw_psl.1.0.0.UPF

Calculations

Waiting x 5

Name	Formula	Script
HT: H2	H2	pw.x
HT: H2O	H2O1	pw.x
HT: CO2	C1O2	pw.x
HT: CO	C1O1	pw.x
HT: COOH	C1O2H1	pw.x

Structure

Script

Potential

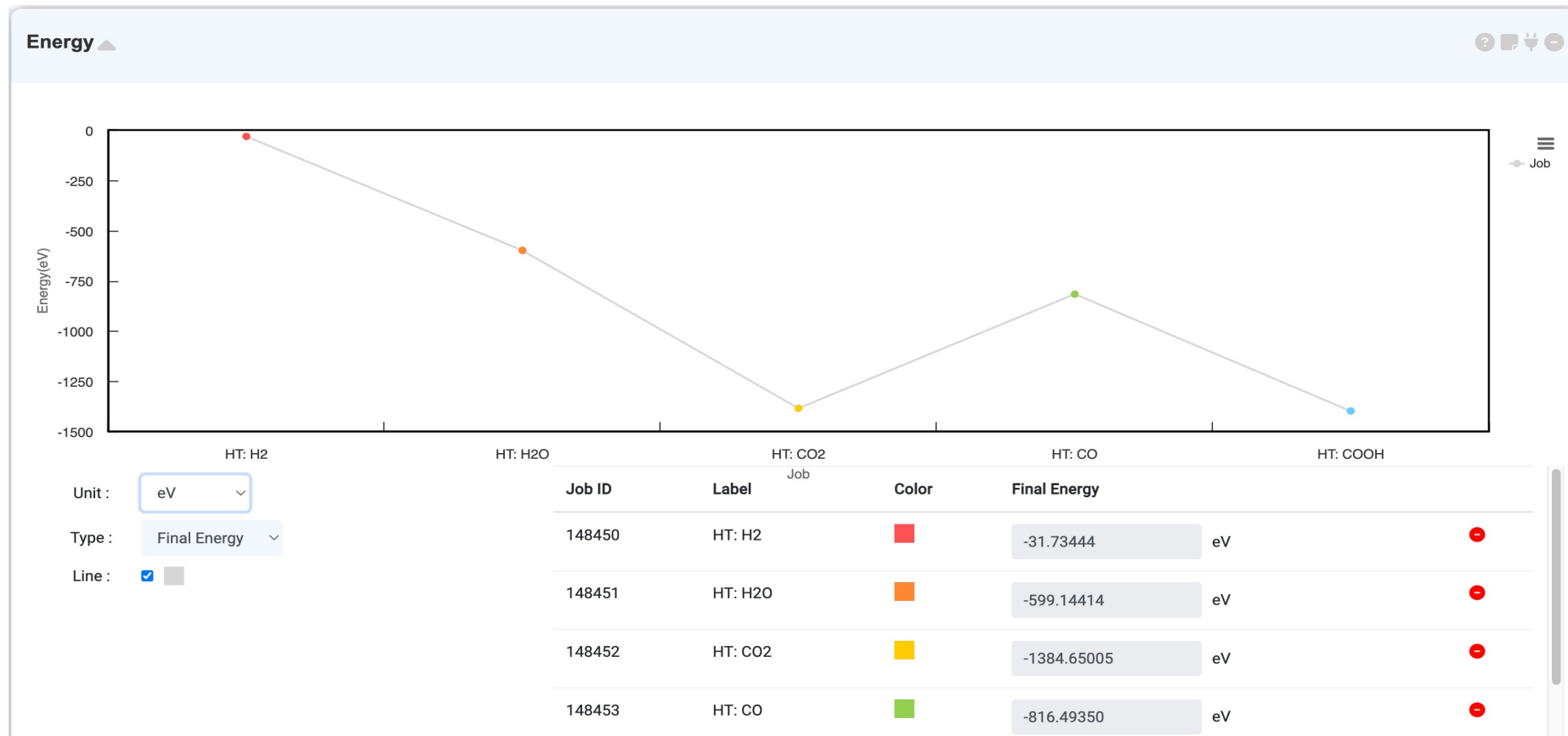
PWscf

```
&CONTROL
calculation = 'relax'
forc_conv_thr = 0.00038
nstep = 200
outdir = './output/'
```

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

10. Calculate adsorption energies

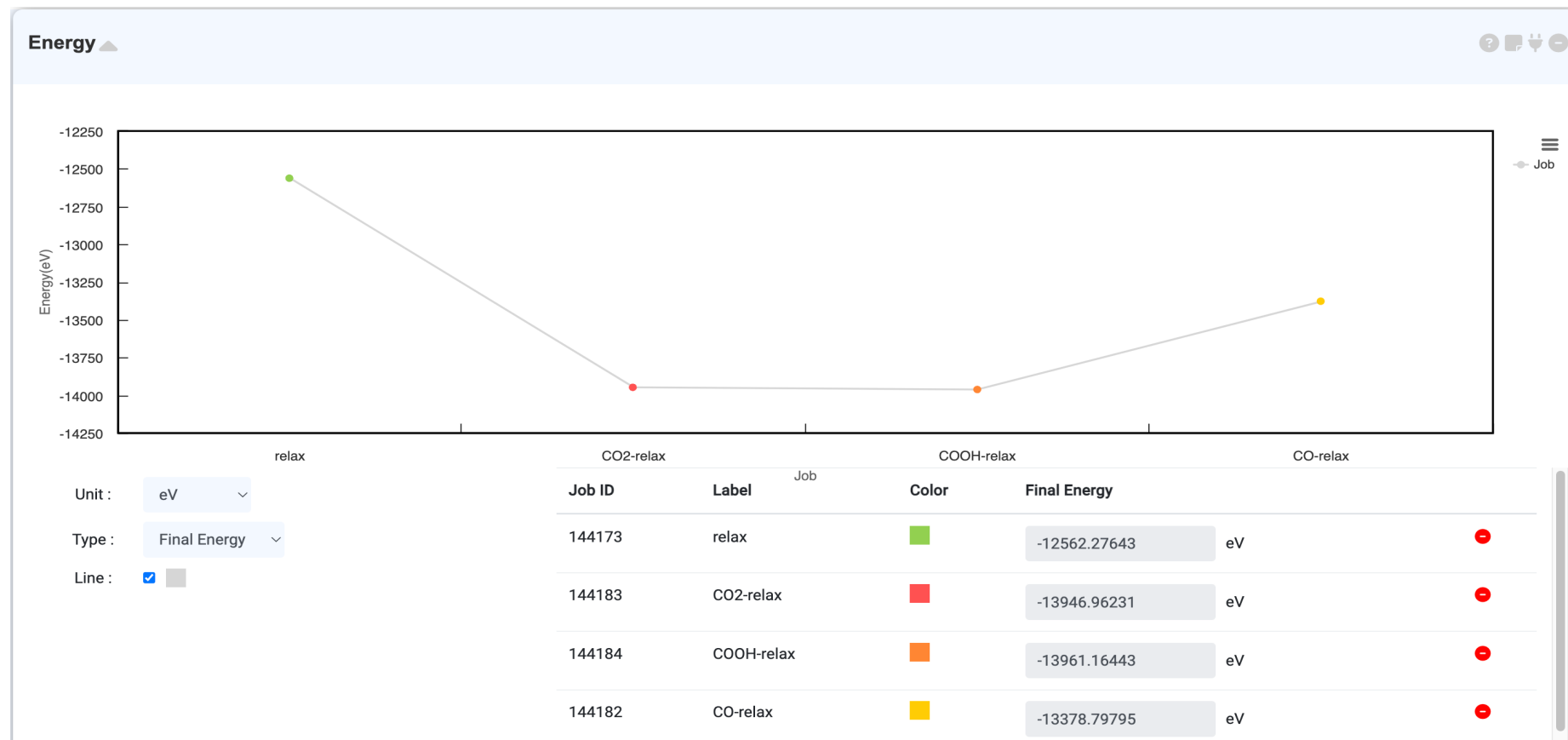
$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{substrate}})]$$



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

10. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{substrate}})]$$



[Lab.4] CO₂ Reduction Reaction (CO₂RR)

10. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{substrate}})]$$

Memo ▲

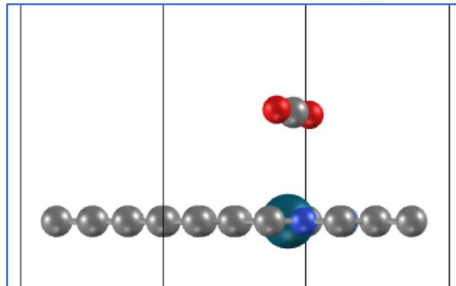
Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼	K ▼	L ▼
1	Structure	Total Energy			Total Energy	Adsorption Energy						
2	H2	-31.73444										
3	H2O	-599.14415										
4	CO2	-1384.6500		CO2/Pd@N4-Gr	-13946.962	0.0358000						
5	COOH*	-1398.4780		COOH/Pd@N4-Gr	-13961.164	0.4099500						
6	CO*	-816.49352		CO/Pd@N4-Gr	-13378.791	=(E6-(B6+B7))						
7	Pd@N4-Gr	-12562.276										
8												
9												
10												

10. Calculate adsorption energies

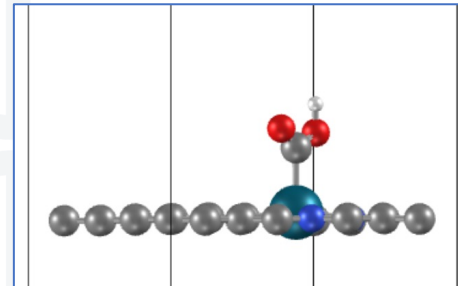
$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{substrate}})]$$

CO₂/Pd@N4-Gr



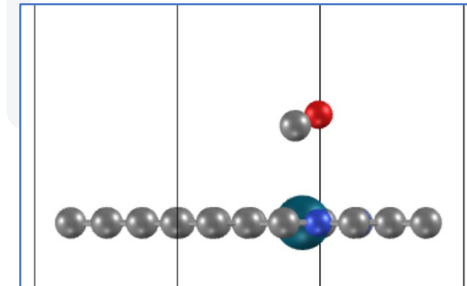
$E_{\text{ads}} = 0.04 \text{ eV}$

COOH/Pd@N4-Gr



$E_{\text{ads}} = 0.41 \text{ eV}$

CO/Pd@N4-Gr



$E_{\text{ads}} = 0.03 \text{ eV}$

[Lab.4] CO₂ Reduction Reaction (CO₂RR)

11. Calculate Free-energy diagram

- $\Delta E_{\text{COOH}^*} = E(\text{COOH}^*/\text{substrate}) - E(\text{substrate}) - \left\{ E(\text{CO}_2) + \frac{1}{2} E(\text{H}_2) \right\}$
- $\Delta E_{\text{CO}^*} = E(\text{CO}^*/\text{substrate}) - E(\text{substrate}) - \{ E(\text{CO}_2) - E(\text{H}_2\text{O}) + E(\text{H}_2) \}$

Memo ▲

Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼	K ▼	L ▼
1	Structure	Total Energy			Total Energy	Adsorption Energy	Delta Energy					
2	H2	-31.73444										
3	H2O	-599.14415										
4	CO2	-1384.6500		CO2/Pd@N	-13946.962	0.0358000	-0.0358000					
5	COOH*	-1398.4780		COOH/Pd@N	-13961.164	0.4099500	1.6292999					
6	CO*	-816.49352		CO/Pd@N	-13378.797	0.0280000	=E6-B7-(B4-B3+B2)					
7	Pd@N4-Gr	-12562.276										
8												
9												
10												

11. Calculate Free-energy diagram

- $\Delta E_{\text{COOH}^*} = E(\text{COOH}^*/\text{substrate}) - E(\text{substrate}) - \left\{ E(\text{CO}_2) + \frac{1}{2} E(\text{H}_2) \right\}$
- $\Delta E_{\text{CO}^*} = E(\text{CO}^*/\text{substrate}) - E(\text{substrate}) - \{ E(\text{CO}_2) - E(\text{H}_2\text{O}) + E(\text{H}_2) \}$

